

WAZUH SIEM LAB Nmap Scan Detection

SOC Analyst Lab Project

Lab Date: 15 May 2026

Report Date: 17 May 2026

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Environment: VirtualBox Lab

Attacker: Kali Linux – 192.168.1.55

Server: CentOS Stream – 192.168.1.59

Tools Used: Wazuh SIEM, Kali Linux, Nmap, CentOS Stream, Windows 11

1. Introduction

1.1 Project Objective

The main objective of this project is to deploy Wazuh SIEM in a home lab environment, connect a Kali Linux agent, simulate a network scan using Nmap, and detect the scan activity through Wazuh security alerts. This project demonstrates real-world SIEM monitoring and alert detection from a Blue Team perspective.

1.2 Scope

This project covers:

- Installing Wazuh SIEM on CentOS Stream VM
- Accessing Wazuh Dashboard via browser
- Deploying Wazuh Agent on Kali Linux VM
- Simulating Nmap network scan from Kali Linux
- Detecting scan activity in Wazuh Security Events
- Documenting Indicators of Compromise (IOCs)

1.3 Tools Used

Tool	Purpose
Wazuh SIEM	Security monitoring and alert detection
CentOS	Stream Wazuh server machine
Kali Linux	Agent machine and attacker
Nmap	Network scanning tool
Virtualbox	Lab environment

2. Lab Setup

2.1 Lab Environment

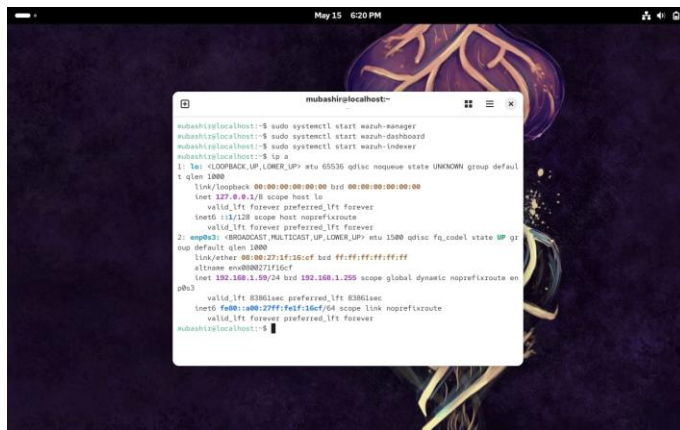
Machine	Operating System	IP Address	Role
Server	CentOS Stream 10	192.168.1.59	Wazuh SIEM Server
Agent	Kali Linux 2025.4	192.168.1.55	Monitored Machine

2.2 Network Setup

Both machines are connected through a private virtual network using VirtualBox Bridge Adapter. This ensures an isolated lab environment for security testing without external interference.

2.3 IP Address Verification

Screenshot 1: CentOS Server IP Address



Note: Kali Linux IP address was 192.168.1.55 as shown in agent dashboard (Screenshot 7)

3. Wazuh Server Installation

3.1 Installation Command

The following command was used to download and install Wazuh SIEM on CentOS Stream:

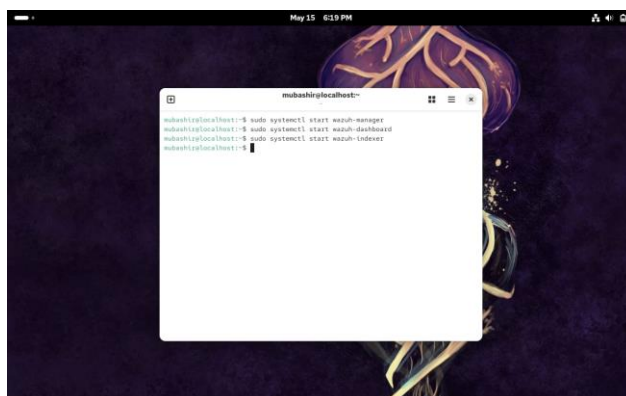
```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh && sudo bash wazuh-install.sh -a -i
```

3.2 Services Started

After installation, the following commands were used to start Wazuh services:

```
sudo systemctl start wazuh-manager  
sudo systemctl start wazuh-dashboard  
sudo systemctl start wazuh-indexer
```

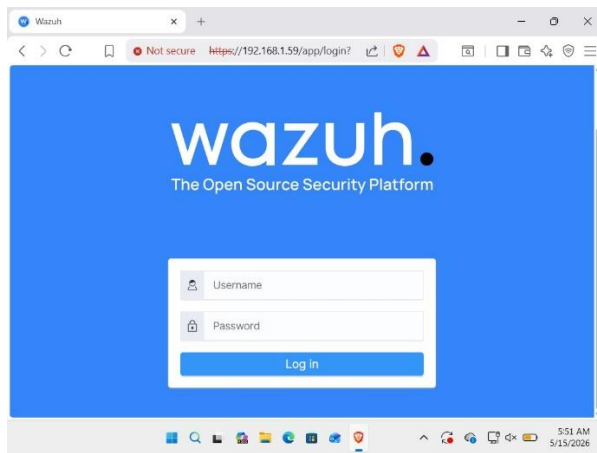
Screenshot 2: Wazuh services scope started on CentOS



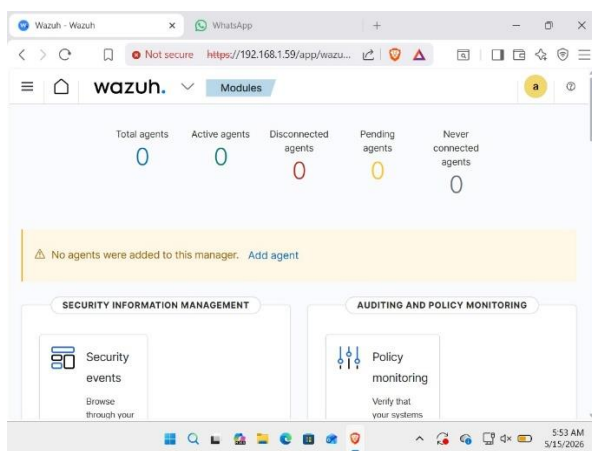
3.3 Dashboard Access

Wazuh dashboard was accessed via browser at <https://192.168.1.59>

Screenshot 3: Wazuh Login Page



Screenshot 4: Wazuh Dashboard Home



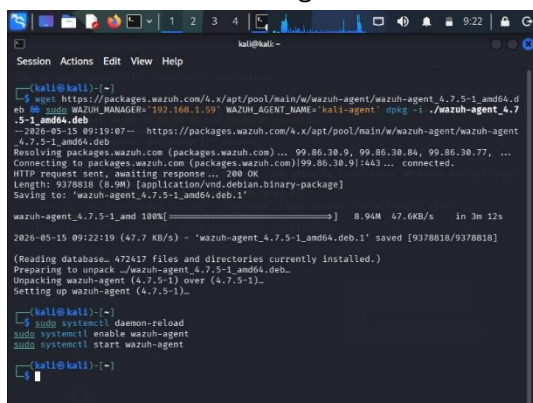
4. Wazuh Agent Deployment on Kali Linux

4.1 Agent Installation

The following command was used to download and install Wazuh Agent on Kali Linux:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb && sudo WAZUH_MANAGER='192.168.1.59' WAZUH_AGENT_NAME='kali-agent' dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb
```

Screenshot 5: Wazuh Agent installation on Kali Linux



4.2 Agent Started

The following commands were used to start the Wazuh Agent:

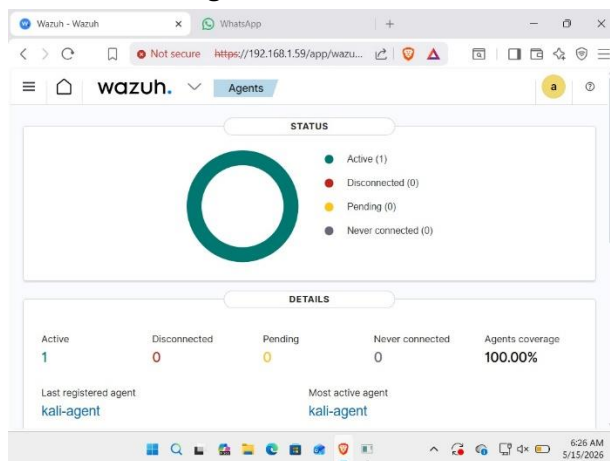
```
sudo systemctl daemon-reload
```

```
sudo systemctl enable wazuh-agent
```

```
sudo systemctl start wazuh-agent
```

4.3 Agent Active in Dashboard

Screenshot 6: Agent Status — Active



Screenshot 7: kali-agent connected in Agents table

The screenshot shows the 'Agents (1)' table in the Wazuh dashboard. The table lists the details of the connected agent, 'kali-agent'.

ID	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	kali-agent	192.168.1.55	default	Kali GNU/Linux 2025.4	node01	v4.7.5	Active	Refresh Export

5. Nmap Scan (Attack Simulation)

5.1 Scan Objective

Nmap was used to perform a TCP connect scan from Kali Linux targeting the Wazuh Server at 192.168.1.59. The purpose was to simulate a real-world reconnaissance attack and verify if Wazuh SIEM detects it.

5.2 Scan Command

Initial scan attempt failed without root privileges. Switched to root user using `sudo su` command, then scan executed successfully.

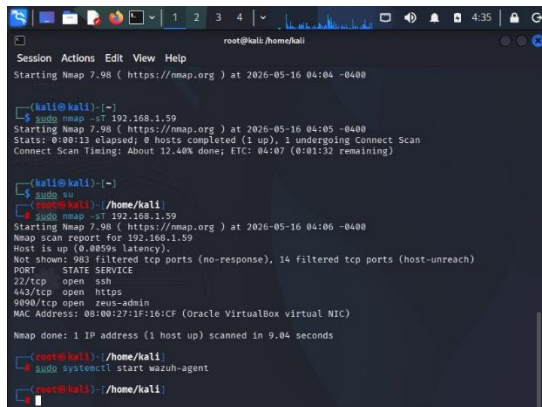
```
sudo su
```

```
nmap -sT 192.168.1.59
```

nmap -sV 192.168.1.59

Flag	Explanation
-sT	TCP Connect Scan — full connection established
-sV	Which version is running on open port
sudo su	Switched to root user for scan execution

Screenshot 8: Nmap scan command and output



```
root@kali: /home/kali
Session Actions Edit View Help
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-16 04:04 -0400

(kali@kali) ~
$ sudo nmap -sT 192.168.1.59
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-16 04:05 -0400
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 12.40% done; ETC: 04:07 (0:01:32 remaining)

(kali@kali) ~
$ sudo su
root@kali: /home/kali
$ sudo nmap -sT 192.168.1.59
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-16 04:06 -0400
Nmap scan report for 192.168.1.59
Host is up (0.0059s latency).
Not shown: 983 filtered tcp ports (no-response), 14 filtered tcp ports (host-unreach)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp   open  https
9890/tcp  open  zeus-admin
MAC Address: 08:00:27:1F:16:CF (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 9.04 seconds

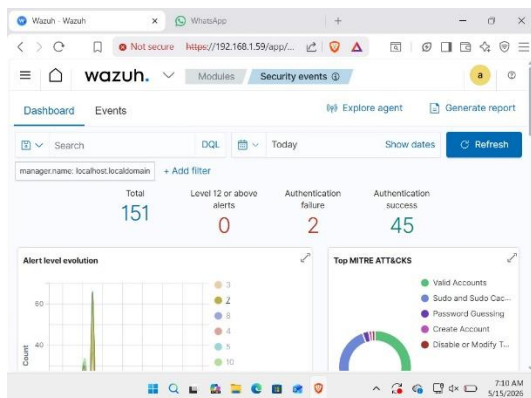
root@kali: /home/kali
$ sudo systemctl start wazuh-agent
$ sudo systemctl start wazuh-agent
$ sudo systemctl start wazuh-agent
```

6. Detection in Wazuh SIEM

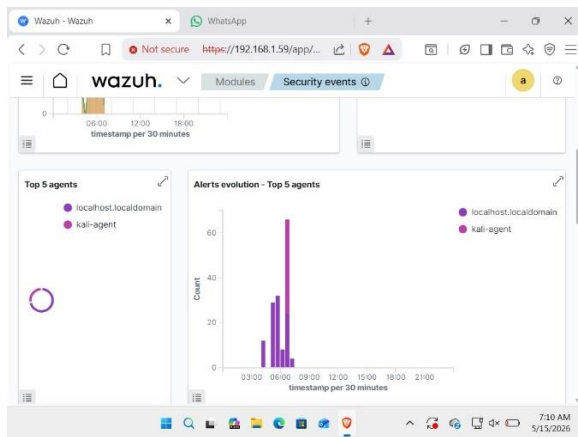
6.1 Security Events Dashboard

After the Nmap scan, Wazuh SIEM successfully generated security alerts for the kali-agent.

Screenshot 9: Security Events Dashboard — Total Alerts

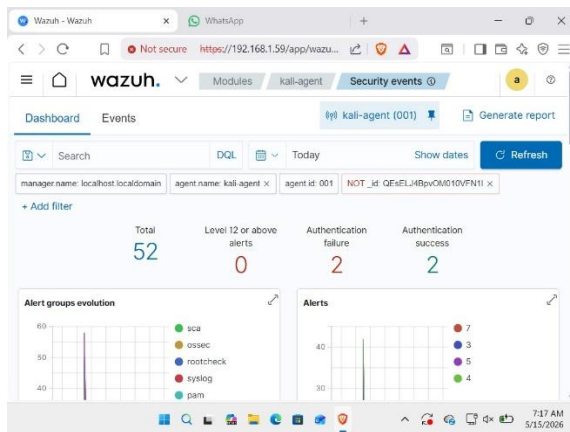


Screenshot 10: kali-agent Alert Evolution



6.2 Alert Analysis

Screenshot 11: kali-agent Security Events Detail



Screenshot 12: Security Alerts List — kali-agent

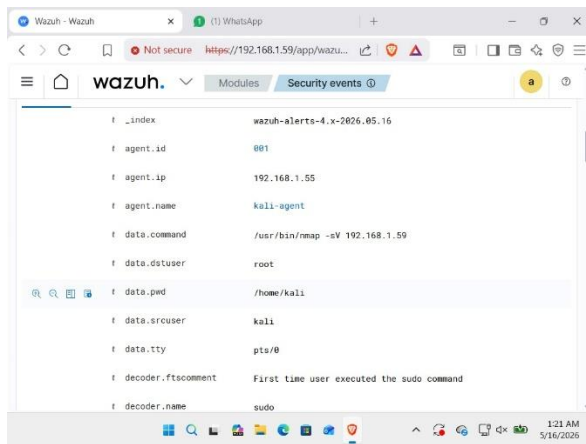
The screenshot shows the Wazuh Security events dashboard for 'kali-agent (001)'. The 'Alerts' tab is active, displaying a list of security alerts. The table has columns for 'Time', 'agent.name', 'rule.description', 'rule.level', and 'rule.id'. The alerts are sorted by time, showing a sequence of login session events and successful sudo commands.

Time	agent.name	rule.description	rule.level	rule.id
May 16, 2025 @ 00:59:38.534	kali-agent	PAM: Login session opened.	3	5501
May 16, 2025 @ 00:59:38.452	kali-agent	Successful sudo to ROOT exe outed.	3	5402
May 16, 2025 @ 00:59:12.404	kali-agent	PAM: Login session closed.	3	5502
May 16, 2025 @ 00:59:16.438	kali-agent	PAM: Login session opened.	3	5501
May 16, 2025 @ 00:59:16.396	kali-agent	Successful sudo to ROOT exe outed.	3	5402
May 16, 2025 @ 00:58:52.979	kali-agent	PAM: Login session closed.	3	5502
May 16, 2025 @ 00:58:52.348	kali-agent	PAM: Login session opened.	3	5501

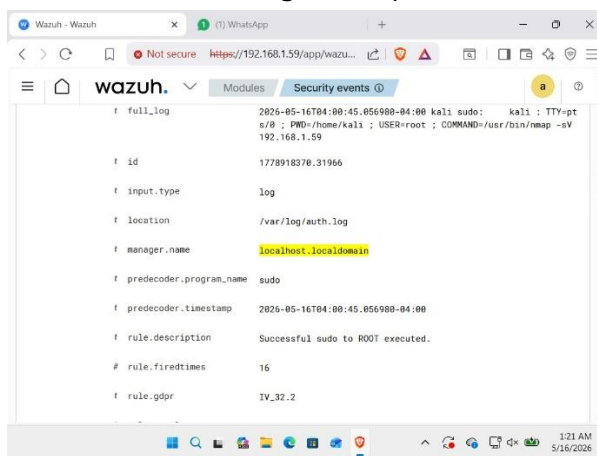
6.3 Nmap Command Detected

Wazuh successfully logged the Nmap scan execution from kali-agent.

Screenshot 13: Nmap command detected in Wazuh



Screenshot 14: Full log of Nmap execution



7. Indicators of Compromise (IOCs)

Definition

Indicators of Compromise (IOCs) are forensic artifacts or pieces of evidence observed on a network or system that indicate potential malicious activity. These artifacts help security analysts detect, investigate, and respond to cyber threats.

7.1 IOCs Identified in this Project

IOC NO.	Category	Indicator	Description
IOC 01	Network	Nmap scan from 192.168.1.55	Port scanning activity detected on Wazuh Server
IOC 02	Process	/usr/bin/nmap -sT 192.168.1.59 and /usr/bin/nmap -sV	Nmap command executed with root privileges
IOC 03	Behavioural	sudo su executed before scan	Privilege escalation before reconnaissance
IOC 04	Network	Port 22, 443, 9090 open	Open ports identified on Wazuh Server

7.2 IOC Severity Classification

IOC NO.	Severity	Justification
IOC 01	High	Port scanning is a common first step in attacks
IOC 02	High	Nmap execution logged by Wazuh SIEM
IOC 03	Medium	Privilege escalation indicates suspicious behavior
IOC 04	Low	Open ports identified during reconnaissance

8. Conclusion

8.1 Project Summary

This project successfully demonstrated the deployment of Wazuh SIEM in a home lab environment, followed by agent deployment on Kali Linux, Nmap scan simulation, and detection of the scan activity through Wazuh security alerts. The entire lab was executed in an isolated VirtualBox environment using CentOS Stream as the Wazuh Server and Kali Linux as the monitored agent machine.

8.2 Key Achievements

- Successfully installed Wazuh SIEM on CentOS Stream
- Deployed Wazuh Agent on Kali Linux and connected to server
- Performed Nmap TCP scan simulating reconnaissance attack
- Wazuh SIEM successfully detected and logged the scan
- Identified 4 Indicators of Compromise (IOCs)
- Demonstrated real-world Blue Team SIEM monitoring

8.3 Learning Outcomes

Skill Area	What I Learned
SIEM Deployment	How to install and configure Wazuh SIEM
Agent Management	How to deploy and connect agents to SIEM
Network Scanning	How Nmap reconnaissance is detected by SIEM
Alert Analysis	How to analyze security events in Wazuh dashboard
IOC Identification	How to document forensic evidence from SIEM alerts

8.4 Challenges Faced & Solutions

Challenge	Solution
Ubuntu Server version not supported	Switched to CentOS Stream
Wazuh Agent architecture mismatch	Downloaded correct amd64 package
Agent not connecting to server	Opened firewall ports 1514 and 1515
Nmap not running without root	Used sudo su to switch to root user

8.5 Future Scope

This project can be extended by:

- Simulating brute force attacks and detecting them in Wazuh
- Adding Windows machine as second agent
- Creating custom Wazuh detection rules

- Integrating with Splunk for advanced correlation

9. GitHub Repository

[<https://github.com/Mubashir-soc-01/SOC-Project>]

10. LinkedIn

[<https://www.linkedin.com/in/mohammad-mubashir-b729643b7/>]

THANK YOU

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Date: 17 May 2026